Introduction to Formal Proof
Synopsis of Materials

Bernard Sufrin

Trinity Term, 2021

Abstract

This is a synopsis of the headings of the slides/notes from which the material presented in our ten lectures will be selected.

There is probably more material than can be covered properly in ten lectures without rushing. The extra material is intended to provide extra reading for those interested in further study of logic and formal proof.

It should be clear from the syllabus and the two tutorial sheets which of the material is **examinable**, and the style in which it will be examined.

0: Introduction and Overview

Preliminary remarks
- Course Overview
- Case Study: three presentations of an equational proof
- Stylized presentation with compressed transitive rules
- Inference tree presentation
- Linearized presentation

1: Formal Proofs in Propositional Calculus

Propositional Calculus
- Introduction
- Propositional Language: propositions
- Propositional Language: atomic propositions
- Symbolic representation
- Composing Propositions with Logical Connectives
- Parsing

Natural Deduction in the Propositional Calculus
- Presenting a conjecture
- What is the nature of a valid conjecture?
- What is the purpose of a proof system?
- Proof Rules for conjunction
- Proof Rules for disjunction
- Proof rules as "conjecture transformers"
- Proof Rules for Implication
- A Paradox?
- Rules for iff
- Proof Rules for Negation
- Derived Rules
- A first glance at soundness and completeness

2: Proofs *about* Propositional Calculus

Road Map

Propositional Semantics
- Propositional semantics
- Propositions are a recursive data type
- Proving things about Propositions
- Evaluating propositional formulae
- A lemma about irrelevant atoms
- Definitions: tautology, satisfiability, entailment
- Detour: Tautology and Satisfiability Checking

Soundness of Natural Deduction
- Soundness 1: Definition
- Soundness 2: Proofs represented as data structures
- Soundness 3: a proof checker

Soundness 4: some observations about subproofs
Soundness 6: proof of soundness
Consequence of Soundness
Completeness of Natural Deduction
Statement of the Completeness Theorem for Natural Deduction
Reusing proofs and using proofs-about-proofs
ASIDE: completeness steps in proofs of admissibility
From Natural Deduction to Sequent Calculi
Reformulating ND as a single-conclusion sequent calculus
Epilogue
Is it essential to represent proofs in Haskell?
An alternative approach: valid proofs as a data type
Proof procedures and completeness

3: Predicate Logic (Semantics)

Introduction: Predicate Language
Propositional logic has limits
Informal predicate language: variables, predicates, quantifiers
Informal predicate language: functions
Inference systems for predicate logic
Predicate Calculus Semantics
Formal predicate language: grammar
Free Variables
Substitution
Models and Meanings
The 7PM Model
Evaluation of formulae without variables
Evaluation of Quantified Formulae in the 7PM model
Formalizing "associate with"
Evaluation rules for formulae in a model M with domain Dom
Semantic Entailment
Satisfiability
ASIDE: Partial Functions

4: Formal Proofs in Predicate Logic

Proof Rules for Predicate Calculus
- Predicate Calculus Proofs
- Proof Rules for the logical connectives
- Proof Rules for Quantifiers: ∀-elimination
- Proof Rules for Quantifiers: ∃-introduction
- Proof Rules for Quantifiers: ∃-elimination
- Proof Rules for Quantifiers: ∀-introduction
- Freshness is important
- Summary of the Quantifier Rules
- Proof Rules for Equality
- Derived consequences of substitutivity
- Supplement (May 2018): Empty Domains and Miracles

5: Theories

Extending Predicate Logic
- Theories
- Example: elementary group theory
- Example: theory of Natural Numbers

Theories with several types
- Theories with several types
- Example: typed theory of natural numbers
- Example: typed theory of number lists

Indispensability of the logical quantifiers
- Formal treatment of generalised induction hypotheses